DSA 101 - questions not covered

Problems from the curriculum sheet that do not become their own challenge in the Classwork or Homework contest, and why.

Summary

	In sheet	Codeable	Own challenge	No match	Not coding	Shares one
Classwork	179	164	105	37	15	22
Homework	120	111	71	33	9	7
Total	**299**	**275**	**176**	**70**	**24**	**29**

So 176 of the 299 problems in the sheet get their own challenge. The remaining 123 are listed in sections 1 to 3: 70 with no match, 24 that are not coding problems, and 29 that share a challenge with an earlier problem. Section 4 covers a different risk: 0 problems that *were* added but whose match scored below 0.50 and is probably the wrong problem.

How the matching worked, and what that means for this list

Each problem was matched by comparing its wording against the 1,407 challenges this account has authored and HackerRank's 1,993 public ones. That is a text-similarity guess, not an understanding of the problem, so a close score is evidence and not proof. Sections 1 to 3 are reliable - they record what is structurally absent. Section 4 is the judgement call.

1 No matching challenge found

Nothing in the searched libraries resembles these, so each needs a challenge authored by hand on HackerRank - statement, test cases and a reference solution - before it can appear in a contest. This is the shortest list and the only one that requires real authoring work.

Classwork - 37 problem(s)

ID	Day	Problem
`CW-D01-2`	Day 1	Print your name, class and roll on separate lines
`CW-D01-3`	Day 1	Print a small ASCII box
`CW-D02-1`	Day 2	Declare and print a variable of each type
`CW-D04-3`	Day 4	Read 3 marks, print total and average
`CW-D06-2`	Day 6	Eligible to vote
`CW-D07-2`	Day 7	Grade from marks (A/B/C/F)
`CW-D08-3`	Day 8	Is a point inside a rectangle?
`CW-D09-2`	Day 9	Day of the week from a number
`CW-D10-2`	Day 10	Convert case using ASCII

ID	Day	Problem
CW-D14-2	Day 14	Tables 1..10 with nested loops
CW-D16-1	Day 16	Solid square of stars
CW-D17-1	Day 17	Pyramid of stars
CW-D17-2	Day 17	Inverted pyramid
CW-D18-1	Day 18	Full diamond of stars
CW-D18-2	Day 18	Hollow triangle
CW-D22-1	Day 22	isPrime(n)
CW-D23-1	Day 23	nCr using a factorial helper
CW-D23-3	Day 23	Area functions for shapes
CW-D25-1	Day 25	isPrime optimized
CW-D26-1	Day 26	GCD using Euclid
CW-D27-2	Day 27	Print 1..N and N..1 recursively
CW-D29-1	Day 29	Read N values and print them
CW-D29-2	Day 29	Print in reverse order by index
CW-D30-3	Day 30	Count elements above the average
CW-D33-2	Day 33	Sort vector of pairs by the second value
CW-D33-3	Day 33	Sort strings by length
CW-D34-1	Day 34	Linear search - return the index
CW-D40-2	Day 40	Track the start and end indices of max sum
CW-D42-3	Day 42	Counting sort a small-range array
CW-D43-1	Day 43	Reverse using two pointers
CW-D44-3	Day 44	Move negatives to the left
CW-D45-1	Day 45	LC15 3Sum
CW-D50-3	Day 50	Sum of both diagonals
CW-D52-2	Day 52	Check if a number is odd/even using Bitwise AND
CW-D54-3	Day 54	Generate all subsets using bit manipulation (GfG)
CW-D55-3	Day 55	Check if any subarray has sum 0 (GfG)
CW-D58-3	Day 58	Find row with max 1s (GfG)

Homework - 33 problem(s)

ID	Day	Problem
HW-D01-1	Day 1	Print your daily timetable as text
HW-D01-2	Day 1	Print a 4-line ASCII figure
HW-D02-1	Day 2	Print sizeof of each type
HW-D03-1	Day 3	Total seconds to h:m:s

ID	Day	Problem
HW-D04-1	Day 4	Bill = price x quantity (2 decimals)
HW-D05-1	Day 5	Voting eligibility as a boolean
HW-D07-1	Day 7	Largest of three distinct numbers
HW-D08-1	Day 8	Ticket price by age group
HW-D09-2	Day 9	Number of days in a month
HW-D13-1	Day 13	GCD by repeated subtraction
HW-D14-2	Day 14	Print a coordinate grid
HW-D15-1	Day 15	Arithmetic-progression sum
HW-D16-1	Day 16	Floyd's number triangle
HW-D22-1	Day 22	sumOfDigits(n)
HW-D22-2	Day 22	isPalindromeNumber(n)
HW-D23-2	Day 23	isLeapYear(y) returning bool
HW-D25-1	Day 25	Sieve of Eratosthenes
HW-D26-2	Day 26	Reduce a fraction to lowest terms
HW-D31-1	Day 31	Count total swaps in bubble sort
HW-D32-1	Day 32	Sort an array of strings by first letter
HW-D34-1	Day 34	Check if a value exists (boolean)
HW-D38-2	Day 38	Rearrange positives and negatives alternately
HW-D39-2	Day 39	Range increment with a difference array (GfG)
HW-D41-2	Day 41	Element appearing once
HW-D42-2	Day 42	Check if arrays are permutations of each other
HW-D44-2	Day 44	Container with most water (intro)
HW-D45-1	Day 45	LC16 3Sum Closest
HW-D46-1	Day 46	LC1343 Number of Subarrays of Size K and Average Greater than or Equal to Threshold
HW-D48-2	Day 48	Longest word in a sentence
HW-D50-2	Day 50	Check the identity matrix
HW-D52-1	Day 52	Multiply/Divide a number by 2 using bit shifts
HW-D54-2	Day 54	Find the two non-repeating elements (GfG)
HW-D57-2	Day 57	Check if a string is a valid shuffle of two strings (GfG)

2 Not a coding problem

Paper exercises, readings and analysis tasks. No judge can grade these, so they are excluded from the contests by design and belong in class discussion or a worksheet instead. Nothing to fix here - this section is for completeness, so you can confirm nothing was dropped by mistake.

Classwork - 15 problem(s)

ID	Day	Problem
CW-D03-3	Day 3	Evaluate a mixed expression, then verify
CW-D19-1	Day 19	Calculate TC of a standard for loop
CW-D19-2	Day 19	Calculate TC of a while loop halving N
CW-D19-3	Day 19	Differentiate O(1) ops from O(N) ops
CW-D20-1	Day 20	TC of nested loop (1 to N, 1 to N)
CW-D20-2	Day 20	TC of dependent loop (i from 1 to N, j from 1 to i)
CW-D20-3	Day 20	TC of i *= 2 loop (Log N)
CW-D21-1	Day 21	Space complexity of creating an array of size N
CW-D21-2	Day 21	Space complexity of basic recursion
CW-D21-3	Day 21	Time-Space trade-off examples
CW-D31-3	Day 31	Trace selection sort on paper
CW-D32-2	Day 32	Trace insertion sort shifts
CW-D32-3	Day 32	Compare stability of Bubble vs Selection
CW-D60-1	Day 60	Mock Test: 3 standard problems (Easy, Med, Med) across Arrays/Strings/Bits
CW-D60-2	Day 60	Dry run space/time analysis of your solutions

Homework - 9 problem(s)

ID	Day	Problem
HW-D05-2	Day 5	Print the truth tables of && and ||
HW-D19-1	Day 19	Read up on Omega and Theta notation
HW-D19-2	Day 19	Determine TC of finding max element in array
HW-D20-1	Day 20	TC of three nested loops (O(N^3))
HW-D20-2	Day 20	TC of loop jumping by i+=2 vs i*=2
HW-D21-1	Day 21	Calculate SC of nested arrays (matrices)
HW-D21-2	Day 21	Find O(1) space solutions vs O(N) solutions
HW-D60-1	Day 60	Complete any pending HWs
HW-D60-2	Day 60	Rest and prepare for DSA 301!

3 Shares a challenge with an earlier problem

These matched a challenge that an earlier problem already claimed, and a contest can hold a challenge only once. Some are deliberate - the curriculum revisits Count digits on Day 12 and again on Day 24, so one challenge covering both is correct. Others are the matcher collapsing two genuinely different exercises onto one challenge; those want a distinct challenge, and the 'Already used by' column shows what to compare against.

Classwork - 22 problem(s)

ID	Problem	Challenge it maps to	Already used by
CW-D08-1	Leap year with branching	is-leap-year-1	CW-D05-2 Leap year as a boolean expression
CW-D10-1	Print the ASCII of a character	print-character-1	CW-D08-2 Classify a character (vowel/consonant/
CW-D13-2	Reverse a number with while	reverse-of-a-number-8	CW-D12-3 Reverse a number
CW-D13-3	Count digits with while	count-number-of-digits-1-2	CW-D12-2 Count digits
CW-D16-2	Right-angled triangle	right-angled-triangle-6	CW-D15-3 Print a simple triangle
CW-D24-1	Reverse a number	reverse-of-a-number-8	CW-D12-3 Reverse a number
CW-D24-2	Sum of digits	sum-of-digits-81-3	CW-D13-1 Sum of digits with while
CW-D24-3	Count digits	count-number-of-digits-1-2	CW-D12-2 Count digits
CW-D26-3	Check if two numbers are coprime	two-numbers-6	CW-D06-3 Maximum of two numbers
CW-D27-1	Factorial recursively	n-factorial-2-1	CW-D22-2 factorial(n)
CW-D27-3	Sum of digits recursively	sum-of-digits-81-3	CW-D13-1 Sum of digits with while
CW-D34-3	First and last index of x	first-and-last-digit-2	CW-D03-1 Last digit and first digit of a number
CW-D36-2	Swap first and last	first-and-last-digit-2	CW-D03-1 Last digit and first digit of a number
CW-D45-3	Count pairs with difference k	count-the-pairs-of-tracks	CW-D14-3 Count all pairs (i, j)
CW-D47-2	Longest subarray with sum <= k	subarray-with-given-sum-8	CW-D40-3 Subarray with largest sum (all negativ
CW-D49-1	LC242 Valid Anagram	valid-anagram-5	CW-D42-1 LC242 Valid Anagram
CW-D49-2	LC387 First Unique Character in a String	first-unique-character-in-a-string-4	CW-D42-2 LC387 First Unique Character in a Stri
CW-D53-2	LC191 Number of 1 Bits	number-finding	CW-D17-3 Number pyramid
CW-D55-2	Find max subarray sum using O(N^2)	max-subarray-sum-5	CW-D47-1 LC209 Minimum Size Subarray Sum
CW-D56-1	LC53 Maximum Subarray (Review)	maximum-subarray-sum	CW-D40-1 LC53 Maximum Subarray
CW-D56-3	LC560 Subarray Sum Equals K (Using prefix sums/hashmap intro)	max-subarray-sum-5	CW-D47-1 LC209 Minimum Size Subarray Sum
CW-D58-2	LC240 Search a 2D Matrix II	search-an-element-in-2-d-matrix	CW-D58-1 LC74 Search a 2D Matrix

Homework - 7 problem(s)

ID	Problem	Challenge it maps to	Already used by
HW-D24-2	Armstrong number; largest digit	armstrong-number-138	HW-D12-2 Check Armstrong number
HW-D28-2	Reverse a string recursively	reverse-a-string-1-4	HW-D10-1 Reverse a string

ID	Problem	Challenge it maps to	Already used by
HW-D43-1	LC977 Squares of a Sorted Array	squares-of-a-sorted-array-2	HW-D35-2 Find Ceil of x in sorted array (GfG)
HW-D43-2	Remove duplicates from a sorted array (LC26)	squares-of-a-sorted-array-2	HW-D35-2 Find Ceil of x in sorted array (GfG)
HW-D48-1	Print all substrings	print-all-substrings	HW-D14-1 Print all pairs with i < j
HW-D54-1	LC268 Missing Number (Using XOR)	missing-number-finder-1	HW-D41-1 Missing number in 1..N
HW-D58-1	Median in a row-wise sorted Matrix (GfG)	print-the-matrix-row-wise	HW-D37-2 Rotate a matrix row

4 Added, but the match is probably wrong

Nothing here. Every remaining mapping scored at least 0.50, so the weak matches an earlier pass produced have been cleared from the markdown and removed from the contests. The cost of that strictness is visible in section 1: the problems that were previously covered by a poor match now count as unmatched instead.

Classwork - 0 problem(s)

None.

Homework - 0 problem(s)

None.